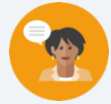


5.6 Using Two- and Three-Column Tables of Equivalent Ratios

Lesson Introduction

 Benchmarks	MA.6.AR.3.3 Extend previous understanding of fractions and numerical patterns to generate or complete a two- or three-column table to display equivalent part-to-part ratios and part-to-part-to-whole ratios. MA.6.AR.3.5 Solve mathematical and real-world problems involving ratios, rates, and unit rates, including comparisons, mixtures, ratios of lengths and conversions within the same measurement system.			 Learning Targets	- I can use two- and three-column tables to display equivalent ratios. - I can solve problems using two- and three-column tables of equivalent ratios.
 Benchmark Coherence	Building On MA.5.AR.3.1 Given a numerical pattern, identify and write a rule that can describe the pattern as an expression.			Working Toward MA.7.AR.3.2 Apply previous understanding of ratios to solve real-world problems involving proportions.	
 Essential Questions	- How can I use two- and three-column tables to display equivalent ratios? - How can I solve problems using two- and three-column tables of equivalent ratios? - When is a two- or three-column table a better representation than a double number line?				
 Lesson Narrative	Students explore equivalent ratios through the lens of two- and three-column tables by comparing and contrasting these models to a double number line. Students should be making connections for the application of multiple representations to display equivalent ratios and solve problems.				
 Component Summary	5.6.1 Warm-Up (~5 min.)	Students consider different ways of representing division (SE p. 198)	5.6.3 Guided Instruction (15-20 min.)	Use three-column ratio tables to display equivalent ratios and solve problems (SE p. 200)	
	5.6.2 Exploration (15 min.)	Students explore the advantages of modeling an equivalent ratio in a two-column table over a double number line (SE p. 199)	5.6.4 Guided Practice (10 min.)	Practice using three-column ratio tables to display equivalent ratios and solve problems (SE p. 202)	
	5.6.5 Wrap-Up (~5 min.)	Formative assessment opportunity using a two-column ratio table to solve real-world rate problems (SE p. 203)			
 Resources	Glossary		Materials		Additional Preparation
	<u>Key Terms:</u> N/A <u>Additional Terms:</u> N/A		(Optional) Note cards (Optional) Chart paper (Optional) Markers		N/A



Teacher Tips

Common Misconceptions

- Students may not independently recognize or understand the advantages of using a two-column table over using a double number line.
- Students may confuse part-to-part and part-to-whole ratios.
- Students may multiply when they should be dividing or divide when they should be multiplying (i.e. students may multiply by 3 in 5.6.3 Guided Instruction question 3a rather than dividing by 3 or multiplying by $\frac{1}{3}$).

Things to Consider

- Explicitly show and explain the features of two- and three- column tables that make them more flexible than double number lines:
 - On a double number line, the differences between numbers are represented by lengths. This can be a limitation when large or small numbers are involved.
 - A double number line dictates the ordering of the values on the line, but in a table, pairs of values can be written in any order.
- Have students draw an arrow from each row of the table with the number by which they need to multiply or divide the numbers in the row to maintain an equivalent ratio. Also, have students write their calculations out so they can check their work. This will help prevent errors.

Literacy and Language Routines

Think-Pair-Share

Consider having students engage in 5.6.3 Guided Instruction using the Think-Pair-Share routine. Initially, have students independently solve the problems. Then, ask students to discuss how they solved the problems and the answer they obtained with a partner. Have students resolve any differences in their answers through discussion. The discussion will help develop student understanding and confidence.

Mathematical Thinking and Reasoning (MTR) Standard

MA.K12.MTR.5.1: Patterns and Structures

The two- and three-column tables help students identify and use the equivalent ratios to solve real-world problems. Students identify the equivalent ratios in the two- and three-column tables. Then, they use the equivalent ratios to solve for unknown values. After solving a few problems, students will develop generalizations based on the similarities found among the problems.



Differentiate Instruction

Have students draw an arrow from each row of the table with the number by which they need to multiply or divide the numbers in the row to maintain an equivalent ratio. For example, notice the table of values in 5.6.3 Guided Instruction. Students may apply rules of distribution, thinking similar strategies would apply. Consider proving both visual and verbal entry points for students to make sense of which values should be multiplied and the reasoning behind the operations.



Cement	Sand	Gravel
1	2	3
$\frac{1}{3}$	$\frac{2}{3}$	1
6.5	13	19.5
5	10	15

Lesson Notes:

5.6.1 Warm-Up: Different Ways of Expressing Division

Component Overview: Students consider different ways division can be represented and explain connections between division and ratios.



5.6.1 Warm-Up: Different Ways of Expressing Division

In many Latin American countries and other countries around the world, the colon (:) is used instead of $\div$ to express division. Therefore, the expression “28 divided by 4” would be expressed as 28:4 instead of $28 \div 4$.

- Using what you have learned about ratios, explain how 28:4 and $28 \div 4$ can be used to express the same value.

Answers will vary.

The ratio 7:1 is equivalent to 28:4. It can be found by dividing both 28 and 4 by 4. 28 divided by 4 is 7.

- Write one question that comes to your mind after reading the information shown.

Answers will vary.

Are there any other symbols that are different in other countries that mean the same thing?



Student answers will vary, and the goal isn't for all students to get this “right.” The purpose is to expose students to differences in the ways people in different parts of the world express mathematics.



After students have written their answers, consider having students Turn and Talk or share whole group other ways they have personally seen division represented, either from personal experience or from others.

5.6.2 Exploration: Double Number Lines Versus Two-Column Tables

Component Overview: Students work with a partner to explore and compare modeling equivalent ratios in two-column tables versus double number lines. Students then have a choice of which representation to use to solve the last problem, explaining their choice in a written explanation.

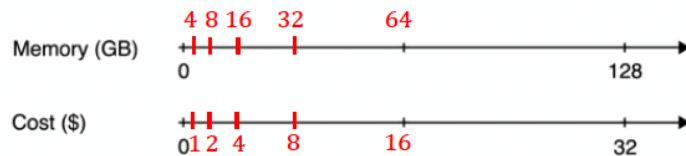


5.6.2 Exploration: Double Number Lines Versus Two-Column Tables

Work with your partner on the following.

In 2016, 128 gigabytes (GB) of portable computer memory cost \$32.

A double number line diagram that represents the situation is shown.



1. One set of tick marks has already been drawn to show the result of multiplying 128 and 32 each by $\frac{1}{2}$. Label the amount of memory and the cost for these tick marks.
2. Next, keep multiplying by $\frac{1}{2}$ and drawing and labeling new tick marks, until you can no longer clearly label each new tick mark with a number.
3. Discuss with your partner whether or not it's possible to clearly represent the cost of 1 GB of memory on the double number line diagram.
4. If it is possible, circle the cost of 1 GB of memory on the double line diagram. If it is not possible, explain why not.

Answers will vary. One would need to circle in between 0 and 1. There isn't enough space to label this clearly on the double line diagram.



Have students intentionally compare and contrast the usefulness of tools like double number lines and tables. When would a double number line be helpful? When would a two-column table be better?

On a double number line, the differences between numbers are represented by lengths. This can be a limitation when large or small numbers are involved. A double number line dictates the ordering of the values on the line, but in a table, pairs of values can be written in any order.

5.6.2 Exploration: Double Number Lines Versus Two-Column Tables (continued)

5. A table representing the same situation is shown. Work with your partner to find the cost of 1 GB. Use as many or few rows as needed.

Memory (gigabytes)	Cost (dollars)
128	32
64	16
32	8
16	4
4	1
1	0.25

Students may choose to use more rows to arrive at this answer. However, if they understand how to find the unit rate at this point (which many will), they may recognize that they can do it in one step by dividing both values by 128.

6. Explain your steps for determining the cost per gigabyte using the table.
Answers will vary. Divide 128 by 2 and until you get to 16 and then begin dividing by 4.
7. Use either the double line diagram or the ratio table to determine the cost of 512 GB of portable computer memory at this rate.

Memory (gigabytes)	Cost (Dollars)
128	32
512	128

Multiply both values by 4. The cost for 512 gigabytes is \$128.



Consider allowing students to share their answers verbally prior to writing their answers. Sharing their answer verbally will allow them to rehearse their explanation and receive feedback. This will improve their confidence and the quality of their responses.



Poll the Class

Consider polling the class to see which students prefer the table and which prefer the double number line. This can lead to a discussion of the advantages of both representations, and the teacher can highlight when it makes the most sense to use each representation.

5.6.2 Exploration: Double Number Lines Versus Two-Column Tables (continued)

8. Discuss with your partner which representation of the equivalent ratios you preferred using to find the costs of 1 GB and 512 GB. Summarize your discussion.

I preferred to use a table because it helps me to visualize the number relationships.



The IBM 7094 was regarded as one of the most powerful advanced mainframe computers in the early 1960s. NASA and the Air Force used it in the Gemini and Apollo space programs.

Despite its large size and historical impact, the IBM 7094 only had a core memory of about 150 kilobytes (KB). That's only enough to manage a few Microsoft Word documents.



Today, you can get a small USB flash drive like the one shown for less than \$20 that can store 4 gigabytes (GB) of memory.



That's almost 27,000 times the memory as the IBM 7094, and you can fit it in your pocket!



Share history or a clip from the movie, *Hidden Figures* to engage more learners in the impact of IBM and important mathematical figures on our technology today.

Mathematician Mary Jackson, pictured below, is one of the key mathematical figures featured in the movie. She and women like her performed all calculations by hand in the early pre-computer days at NASA.



5.6.3 Guided Instruction: Using Three-Column Ratio Tables

Component Overview: Students use a table to find and represent the number of free-throw shots Kiara took at basketball practice. Students find missing values and then explain how they used the table to do so. Students then apply this to a table representing the concrete mixture, using information in the table to find other missing information.



5.6.3 Guided Instruction: Using Three-Column Ratio Tables

Ratios can be used to make part-to-part or part-to-whole comparisons. When comparing parts of a mixture to the whole, three-column ratio tables can be useful to represent the relationships between quantities.

1. Kiara is practicing her free-throw shot at basketball practice. She averages 4 made shots for every 2 missed shots. The table shown can be used to represent this situation.

Number of Shots Made	Number of Shots Missed	Total Shots Taken
4	2	6
16	8	24
2	1	3

- a. When determining this rate during practice, Kiara missed 8 free-throw shots. Complete the second row of the table to show how many shots she made and how many total shots she took.
- b. How was each value in the second row determined?

We are told that she missed 8 shots. Previously she missed 2 shots. Dividing 8 by 2 we obtain 4. This tells us that each value in row one should be multiplied by 4. Multiplying 4 by 4 we see she made 16 shots. Multiplying 6 by 4 we see she took a total of 24 shots. Checking our work, the shots made and shots missed equals the total shots.



Notice the opportunities for verbal mathematical reasoning in this component. Some students may benefit from verbalizing their responses (questions 1b and 2a) before putting them in writing. Ensure that students have an opportunity to engage in discussion to provide this entry point for learners.



Think-Pair-Share

To provide an opportunity for student discussion, engage the class in a “Think-Pair-Share” routine for the first question. Initially, have students independently solve the problems. Then, have students discuss how they solved the problems and the answers they obtained with a partner. Have students resolve any differences in their answers through discussion. As students are discussing, circulate to collect informal data on how students are determining the values in each row. Consider bringing the class back as a group and having students share particularly efficient strategies or draw attention to a common misconception.

5.6.3 Guided Instruction: Using Three-Column Ratio Tables (continued)

2. Kiara had the opportunity to take 3 free-throw shots in her most recent game, and her accuracy remained the same as in practice. Complete the third row of the table to show how many free-throw shots she made and missed in the game.

- a. How was each value in the third row determined?

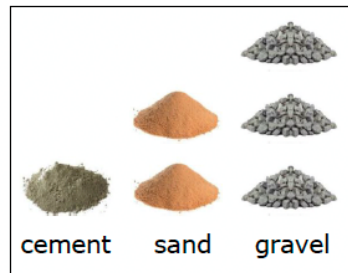
We are told that she took 3 shots. Previously she took 6 shots. Dividing 6 by 3 we obtain 2. This tells us that each value in row one should be divided by 2. Dividing 4 by 2 we see she made 2 shots. Dividing 2 by 2 we see she missed a total of 1 shot. Checking our work, the shots made and shots missed equals the total shots.

- b. Complete the statement.

For every 3 free-throw shots taken, Kiara makes 2 shot(s) and misses 1 shot(s) on average. The ratio of missed shots to made shots to total shots is 1 : 2 : 3.

3. Three-column tables can also be useful when comparing mixtures that contain more than two parts.

Concrete can be created by mixing 1 part cement with 2 parts sand and 3 parts gravel. These dry ingredients are first combined before being mixed with water.



To encourage mathematical reasoning as students fill in the tables, ask questions like:

- Do the answers in your tables make sense? How do you know?
- How did you use the table to help determine what should go in the next row?
- How can you check your work to ensure you filled in the table correctly?

5.6.3 Guided Instruction: Using Three-Column Ratio Tables (continued)

- a. Complete the table to determine equivalent mixtures of concrete. Show how the values in each row are determined using arrows and by indicating the factor used.

Divide by 3
Multiply by 6.5
Multiply by 5



Cement	Sand	Gravel
1	2	3
$\frac{1}{3}$	$\frac{2}{3}$	1
6.5	13	19.5
5	10	15

- b. Marc uses one-gallon buckets to measure out the quantities needed in his concrete mixture. How many gallons of dry mix will he have in one batch of concrete?

Marc will need 6 gallons of dry mix to make one batch of concrete.



Challenge advanced learners to write their own real-world problems that can be displayed and solved using a three-column ratio table. Students can solve their own problems or swap with a partner and solve each other's problems.

5.6.4 Guided Practice: Using Two- and Three-Column Ratio Tables

Component Overview: Students begin this Guided Practice by using a three-column table to find the quantities needed for different paint mixtures, then explain how they completed the table. Students then complete two more tables based on given real world contexts, applying their knowledge of ratios and unit rates to find unknowns in the tables.



5.6.4 Guided Practice: Using Two- and Three-Column Ratio Tables

1. A particular shade of maroon paint calls for 5 cups of red paint to be mixed with 3 cups of blue paint. Complete the table to find the quantities needed for different batches of paint.

Red Paint (cups)	Blue Paint (cups)	Maroon Paint (cups)
5	3	8
20	12	32
1.25	0.75	2
2.5	1.5	4
50	30	80



To provide a visual entry point, have students draw an arrow from each row of the table with the number by which they need to multiply or divide the numbers in the row to maintain an equivalent ratio.

- a. If the mixture in the first row represents one batch of maroon paint, how many batches are in 80 cups of maroon paint?

Ten batches are in 80 cups of maroon paint.

- b. Explain how this relates to the factor used to complete the last row in the table.

Each number of cups in the first row was multiplied by 10 to determine the number of cups in the last row. Therefore, if the first row represents one batch, the last row will represent 10 batches.

5.6.4 Guided Practice: Using Two- and Three-Column Ratio Tables (continued)

2. There are 16 cups in one gallon. Use this information and the table below to determine how many gallons of paint the 80-cup mixture of maroon paint makes.

Cups	Gallons
16	1
80	5

3. In April, Kyla went for a walk on 4 days for every 1 day that she did not walk. There are 30 days in April. Use this information to complete the ratio table.

Walking Days	4	24
Non-Walking Days	1	6
Total Days	5	30

Complete the statement.

Throughout the year, Kyla continued walking 4 days for every 1 day that she did not walk. Based on this, in November, Kyla walked 24 of the 30 days.



After students complete the tables in questions 1, 2 and 3 consider having students share their strategies in small groups. This will provide an opportunity for students to verbalize their thinking and compare their strategies to others' means of solving, allowing them to make connections and/or determine more efficient means of solving.

5.6.5 Wrap-Up: Ticket Out the Door

Component Overview: Students demonstrate their understanding of using two-column ratio tables to solve real-world rate problems by determining the distance traveled when walking at a given rate.



5.6.5 Wrap-Up: Ticket Out the Door

1. Julio participated in a walk-a-thon to raise money for cancer research. Walking at a constant speed, Julio travelled 6.4 miles in the first 2 hours of the walk-a-thon. His goal is to walk for 7 hours. Use the table to find how far Julio will have walked if he reaches his 7-hour goal. Show how the values in each row are determined.

	Hours	Miles Walked	
$\frac{1}{2} \cdot$	2	6.4	$\cdot \frac{1}{2}$
	1	3.2	
$7 \cdot$	7	22.4	$\cdot 7$

Julio will walk 22.4 miles in 7 hours.



Consider using EdgeXL to provide additional enrichment and practice opportunities for students.